



McStas Union components

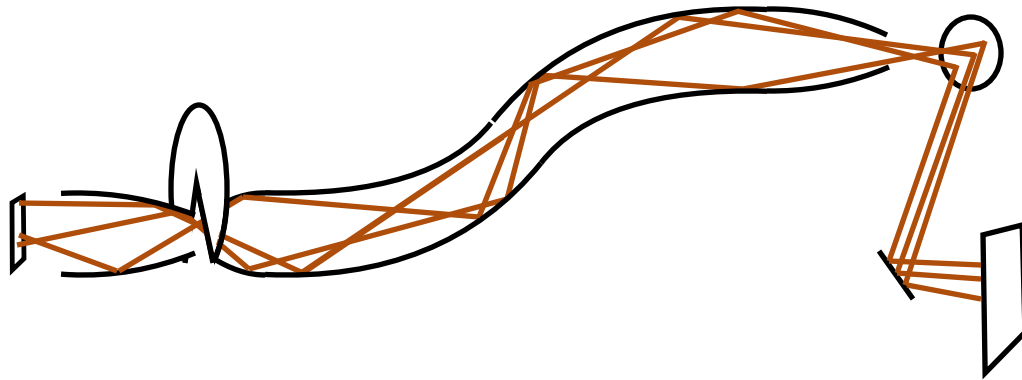
ISIS McStas School 2025

PRESENTED BY MADS BERTELSEN

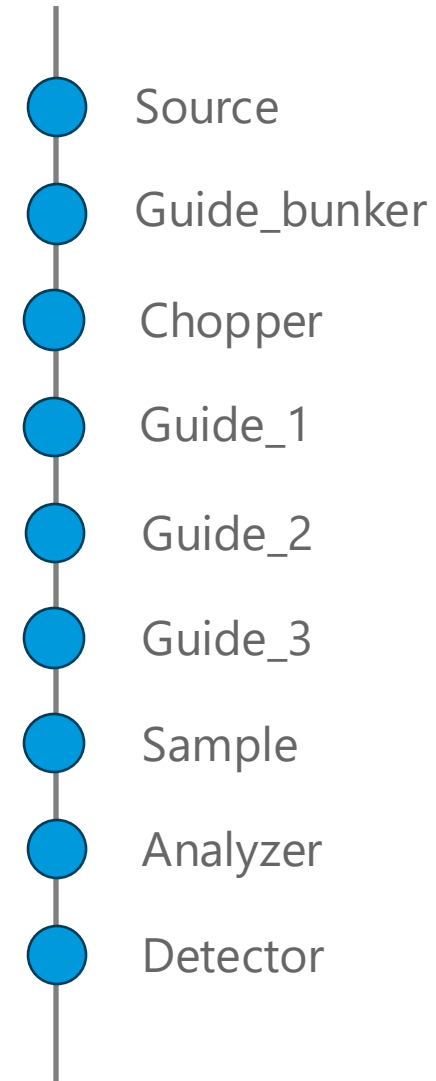
2021-03-03

Introduction

McStas instrument structure



Component sequence



Introduction

Difficulties with realistic samples

- Sample holders with complicated geometry
- Consists of many different materials
- No clear order in which neutrons would pass through
- Could have co aligned crystals
- Twinned crystals
- It could all be in a sample environment

The Union components is one way to overcome these challenges!

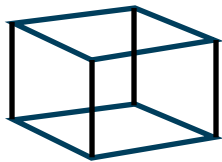


Photo by Pia Ray Jensen

Union components

Using the Union components

Lets simulate a single crystal sample!



Typical McStas sample

Single_crystal.comp

- Geometry
- Raytracing
- Absorption
- Scattering
 - Incoherent
 - Single crystal Bragg
- Multiple scattering loop

Instrument file



Union components

Union_init.comp

Incoherent_process.comp

Incoherent scattering

Single_crystal_process.comp

Single crystal scattering

Union_make_material.comp

Union_box.comp

Union_master.comp

- Raytracing
- Absorption
- Sampling scattering physics
- Multiple scattering

Union_stop.comp

Union components

Modular approach

Think of it this way, its like buying a sofa



Classic McStas sample



Use Union components



Write your own component



Union components

Modular approach



- Union components split the classic sample component into:
 - Process components describing individual scattering processes
 - Materials, collections of processes with additional material constants
 - Geometry components, basic shapes
 - Logger / Absorption logger components, recording debug information
- We will now go through these in turn

Union components

Process components



McStas



- Describe single scattering event
 - Calculates cross section (in inverse penetration depth)
 - Provides function for performing scattering

Process component	Description	Author
Incoherent_process	Incoherent scattering	Converted from classic
Powder_process	Powder Bragg scattering	Converted from classic
Single_crystal_process	Single crystal Bragg	Converted from classic
PhononSimple_process	Phonon branch (transverse or longitudinal)	Converted from classic
NCrystal_process	Full NCrystal material	Converted from classic
AF_HB_1D	Anti-Ferromagnetic Hissenberg chain in 1 dimension	Mads Bertelsen
IncoherentPhonon_process	Phonons from density of states	Victor Laliena
Texture_process	Textured powder	Victor Laliena
Template_process	Starting point when creating a new process	

Physics

Processes

```
COMPONENT Al_incoherent = Incoherent_process(  
  sigma = 4*0.0082, packing_factor=1,  
  unit_cell_volume = 66.4)  
AT (0,0,0) ABSOLUTE
```

```
COMPONENT Al_powder = Powder_process(  
  reflections = "Al.laz")  
AT (0,0,0) ABSOLUTE
```



Processes

Incoherent_process

Powder_process

Physics

Materials



```
COMPONENT Al_incoherent = Incoherent_process(  
  sigma = 4*0.0082, packing_factor=1,  
  unit_cell_volume = 66.4)  
AT (0,0,0) ABSOLUTE
```

```
COMPONENT Al_powder = Powder_process(  
  reflections = "Al.laz")  
AT (0,0,0) ABSOLUTE
```

```
COMPONENT Al = Union_make_material(  
  my_absorption = 100*4*0.231/66.4,  
  process_string = "Al_incoherent,Al_powder",  
  refraction_SLD=2.078E-6)  
AT (0,0,0) ABSOLUTE
```

Processes

Material



Absorption given as inverse penetration depth

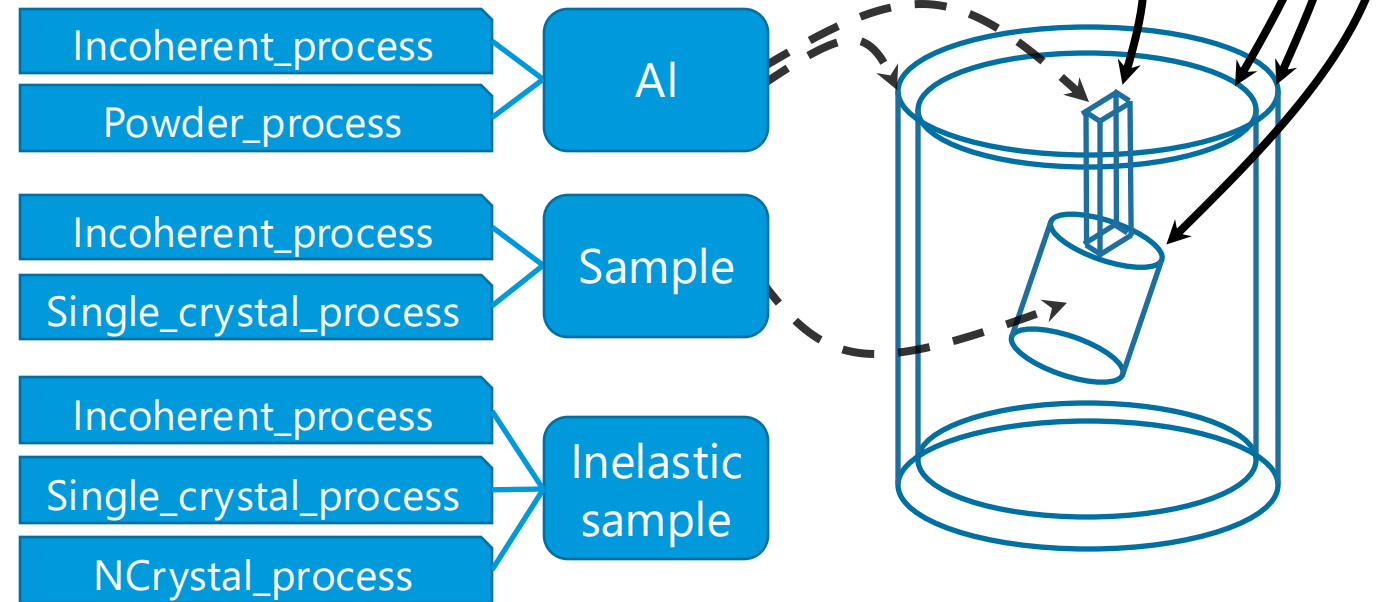
$$\text{my [1/m]} = \frac{\text{cross section per unit cell}}{\text{unit cell volume}}$$

Scattering length density in [$\text{\AA}^{-2}$]

Geometry

Modular geoemtry

- Construct the desired geometry from simple basic shapes
 - Sphere
 - Cylinder
 - Box
 - Cone
 - Mesh model (OFF files in progress)
- Assign each geometry a material





Geometry

Priority

- Each geometry is given a priority
- When two geometries overlap, the overlapping region has the material of the highest priority
- This allows the user to create complex geometries

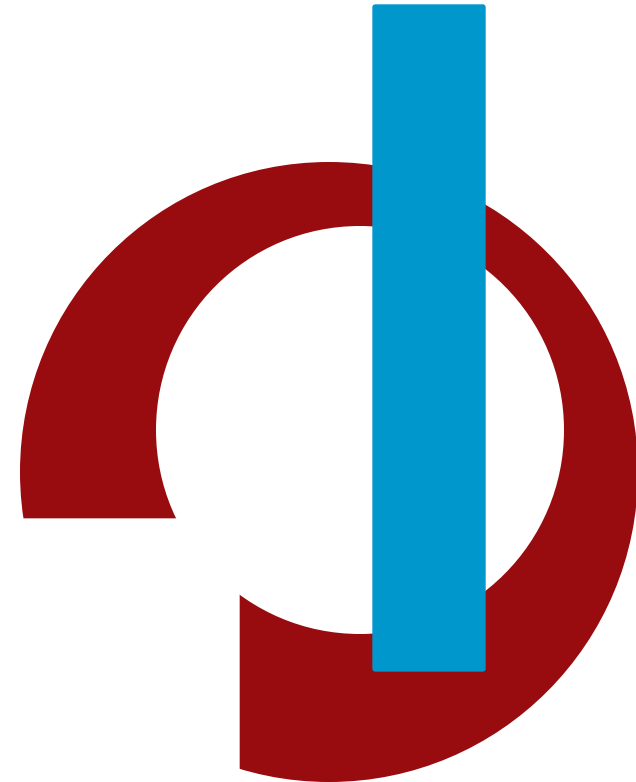
Aluminium
priority = 8.2

Vanadium
priority = 5.1

Geometry

Priority

- Each geometry is given a priority
- When two geometries overlap, the overlapping region has the material of the highest priority
- This allows the user to create complex geometries



Geometry

Modular geometry

```
COMPONENT cryostat_shell = Union_cylinder(  
  radius = 0.15, yheight = 0.4,  
  priority = 10, material_string = "Al")
```

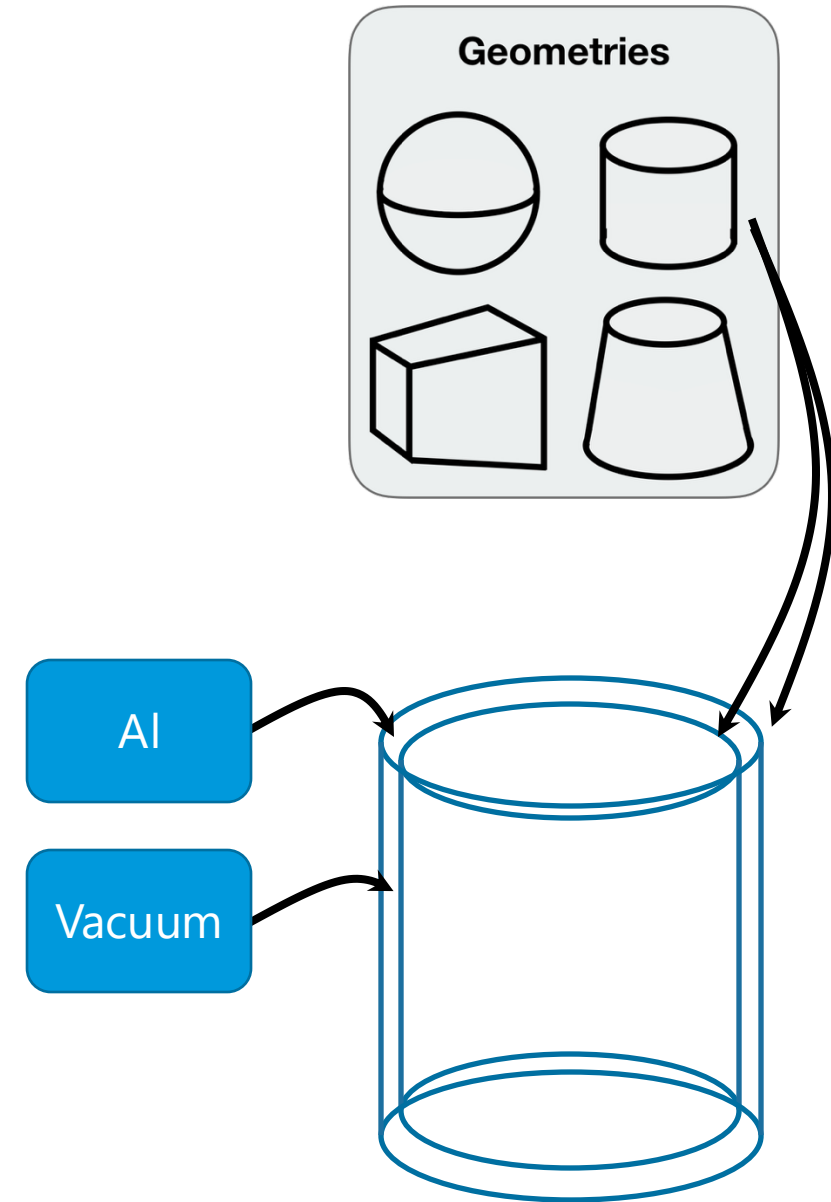
```
AT (0,0,0) RELATIVE target
```

```
ROTATED (0,0,0) RELATIVE target
```

```
COMPONENT cryostat_vacuum = Union_cylinder(  
  radius = 0.147, yheight = 0.39,  
  priority = 11, material_string = "Vacuum")
```

```
AT (0,0,0) RELATIVE target
```

```
ROTATED (0,0,0) RELATIVE target
```



Union master

Where the simulation actually happens

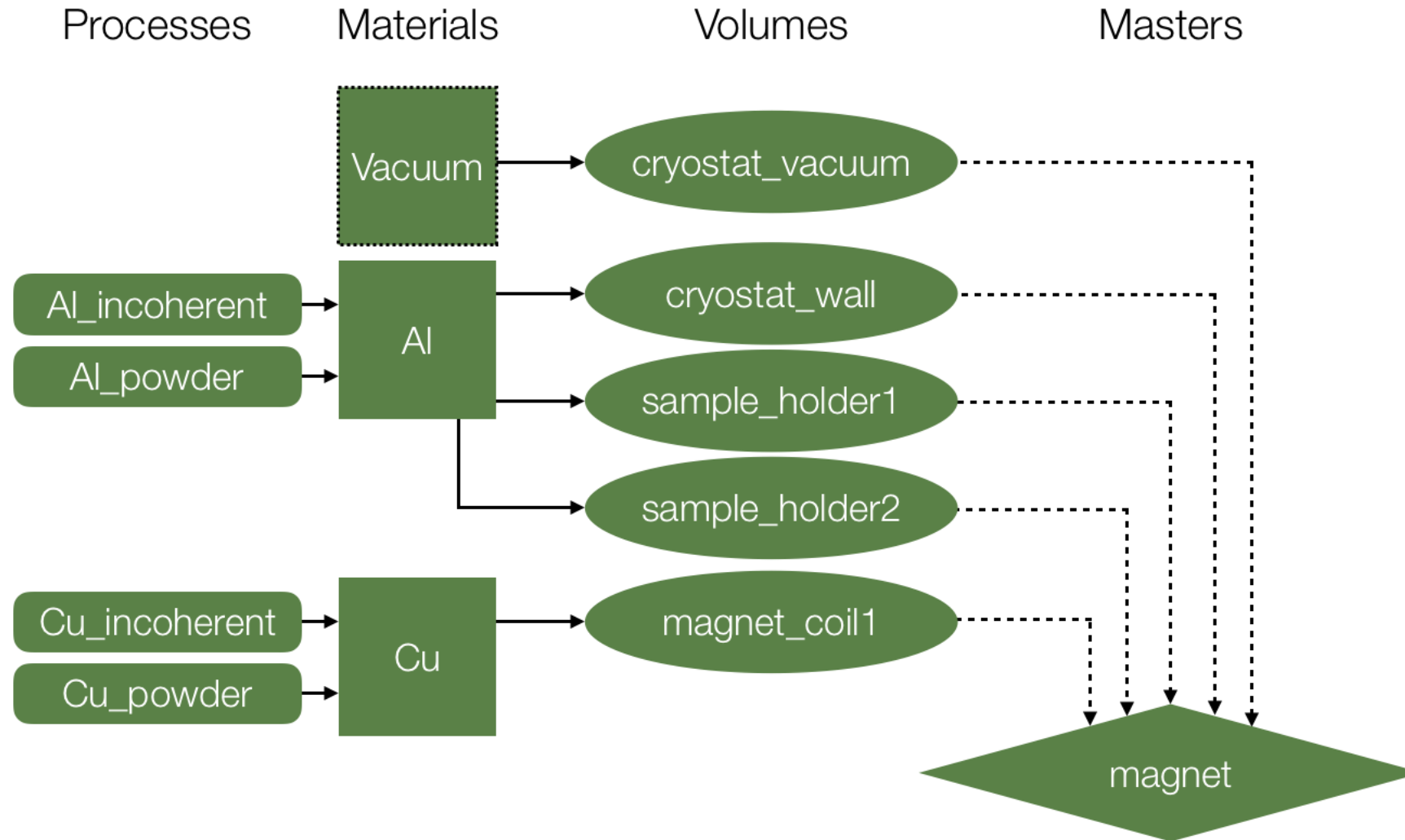


- All components discussed doesn't actually perform any simulation
- Process and geometry components describe the system to be simulated
- The **Union_master** component performs the simulation
- The Union_master component will simulate all Union geometry components defined so far
- No inputs are necessary, as all the information is already in the processes and geometry

```
COMPONENT master = Union_master()  
AT (0,0,0) ABSOLUTE
```

Union master

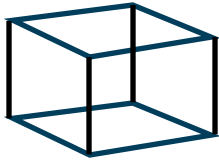
Overview of a Union setup



Union components

Using the Union components

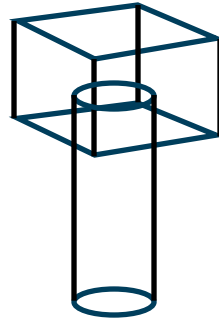
Typical sample component



Single_crystal.comp

Geometry
Raytracing
Absorption
Scattering
- Incoherent
- Single crystal Bragg
Multiple scattering loop

Union components



Union_cylinder.comp

Union_init.comp

Incoherent_process.comp

Incoherent scattering

Single_crystal_process.comp

Single crystal scattering

Union_make_material.comp

Union_box.comp

Union_master.comp

Raytracing
Absorption
Sampling scattering physics
Multiple scattering

Union_stop.comp

Introduction

How does this improve the situation?

- Sample holders with complicated geometry
- Consists of many different materials
- No clear order in which neutrons would pass through
- Could have co aligned crystals
- Twinned crystals
- It could all be in a sample environment

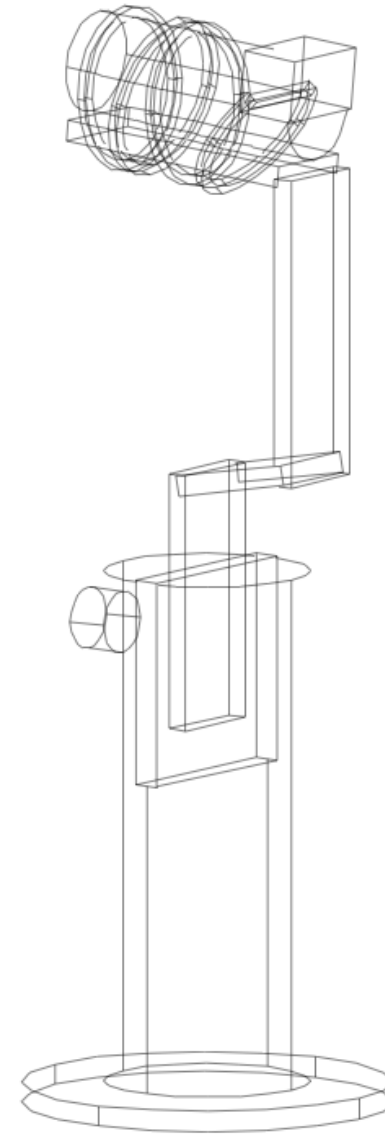
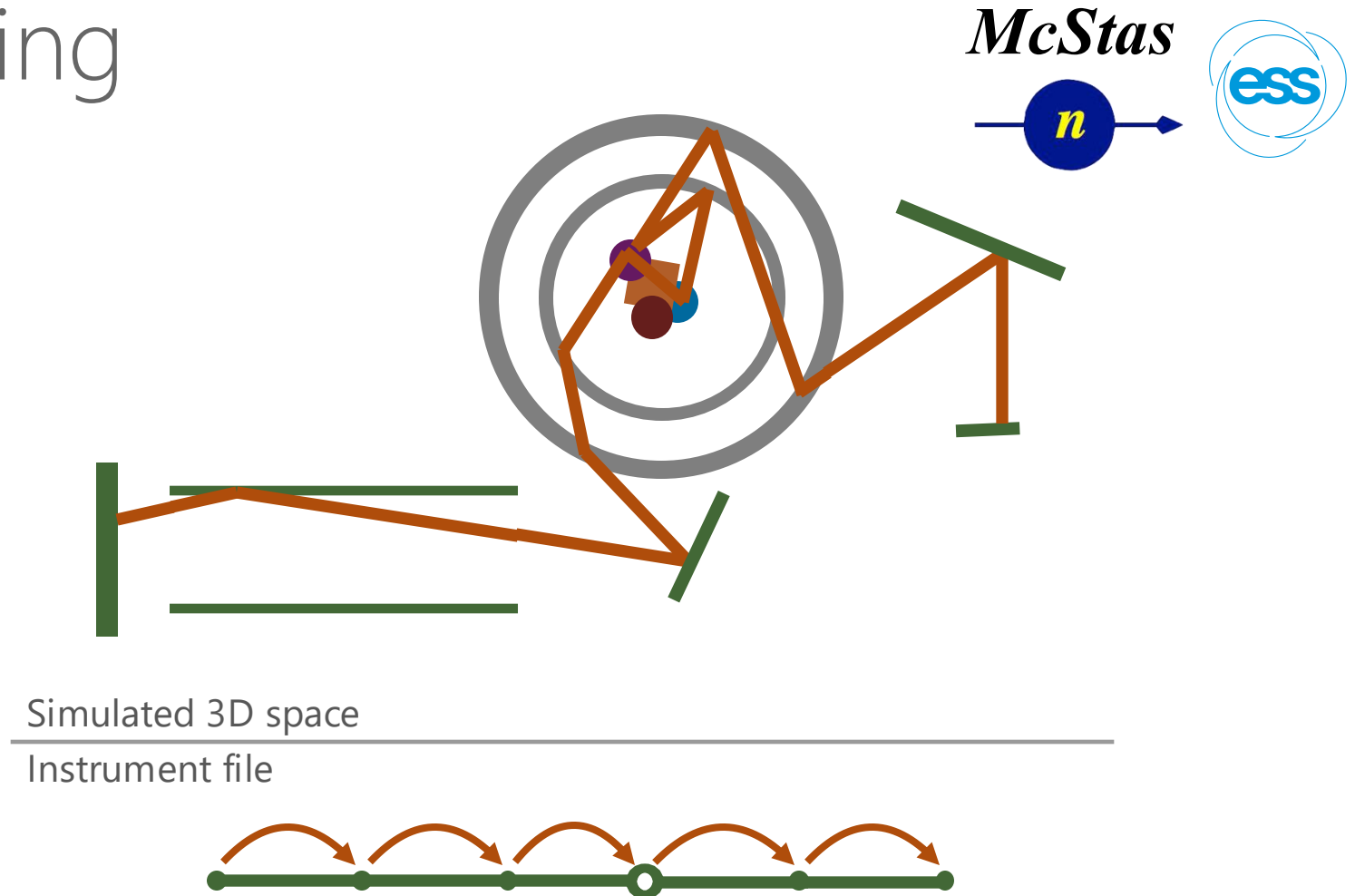


Photo by Pia Ray Jensen

Multiple scattering

Logic in McStas instrument

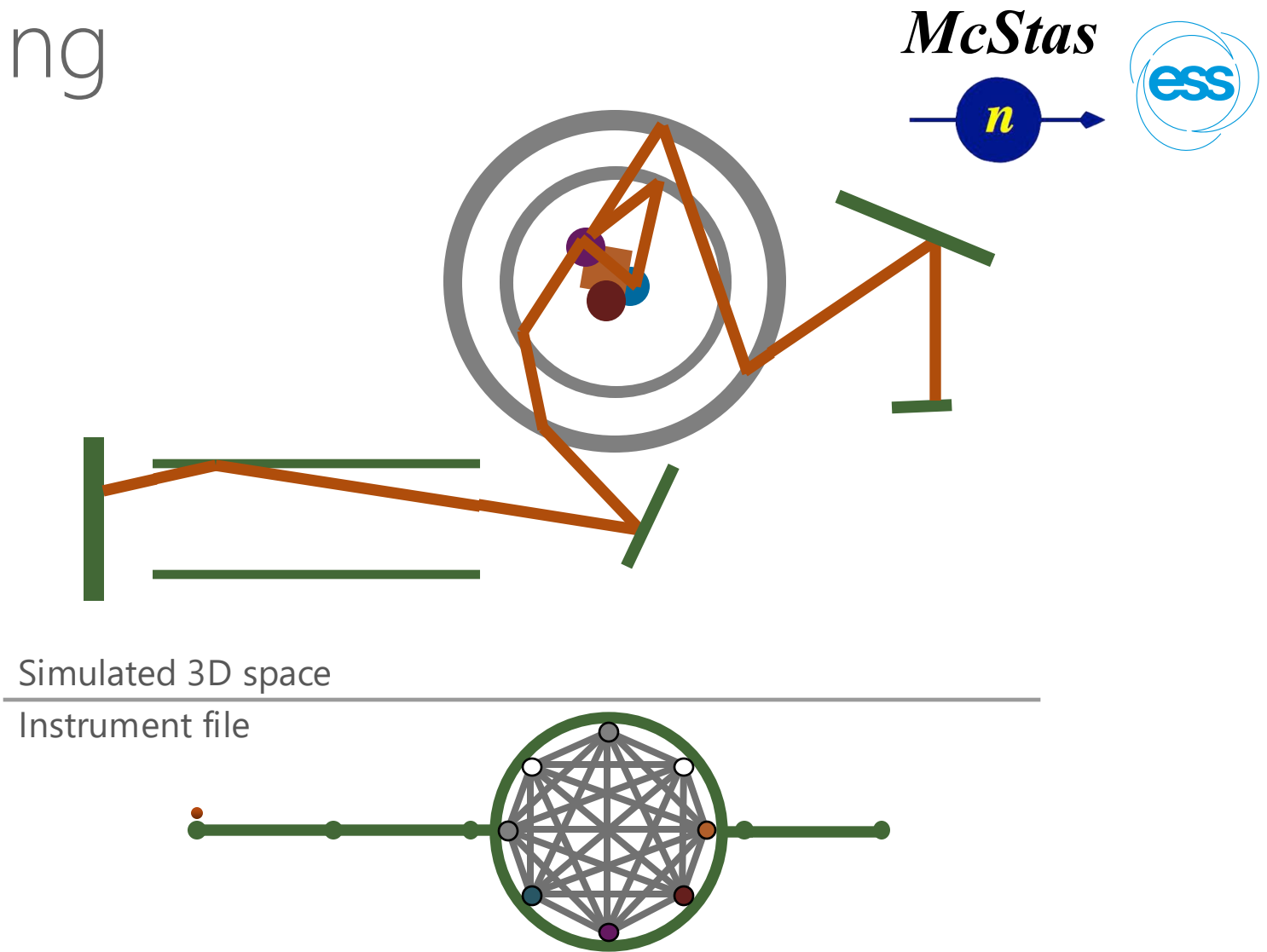
- Multiple scattering handled in Union_master component
- Neutron handed back to instrument component sequence



Multiple scattering

Logic in McStas instrument

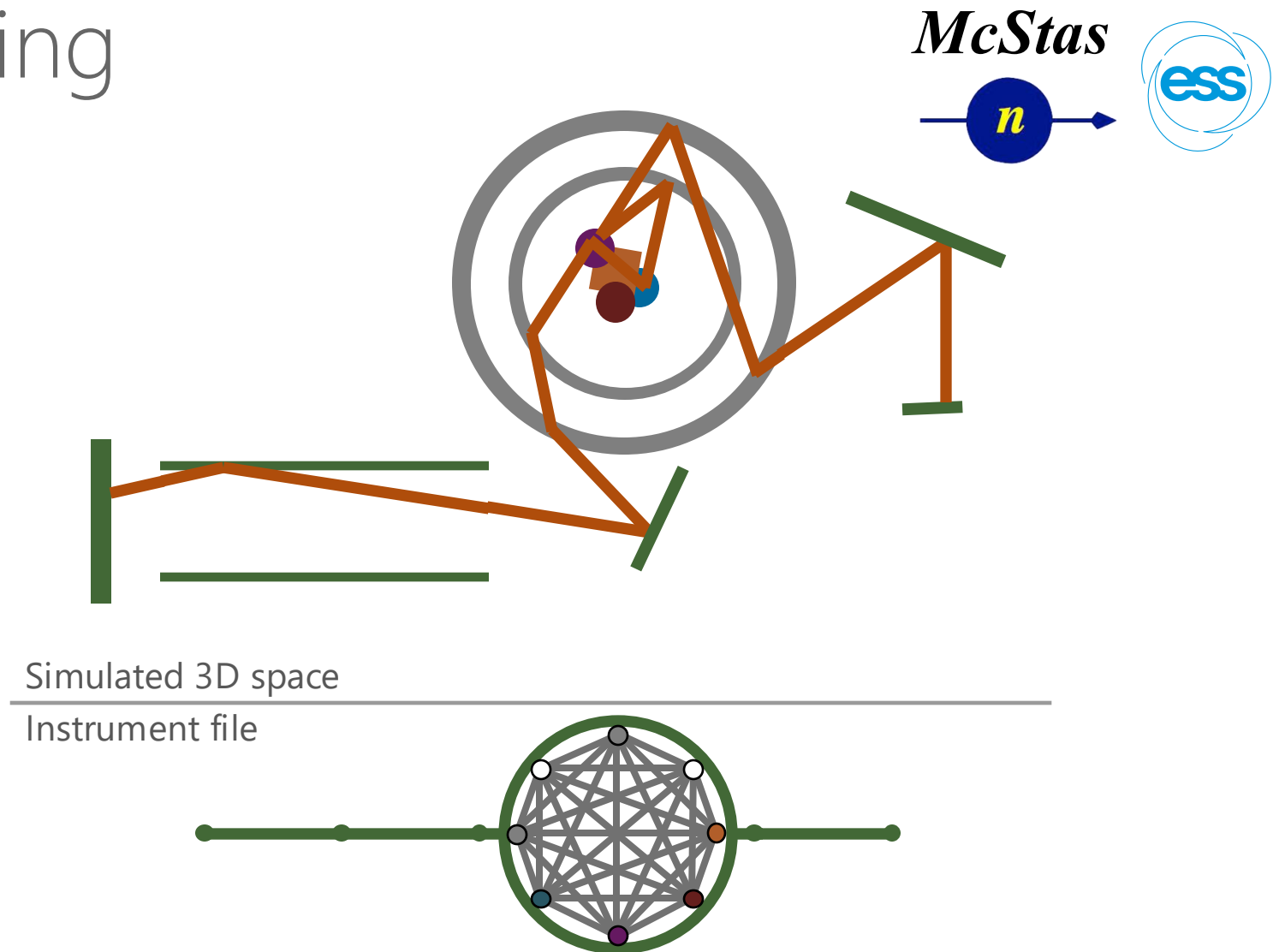
- Multiple scattering handled in Union_master component
- Neutron handed back to instrument component sequence
- Ray can go to geometries in any order



Multiple scattering

Logic in McStas instrument

- Multiple scattering handled in Union_master component
- Neutron handed back to instrument component sequence
- Ray can go to geometries in any order
- Internal logic network reduce number of intersection calculations, improving execution speed



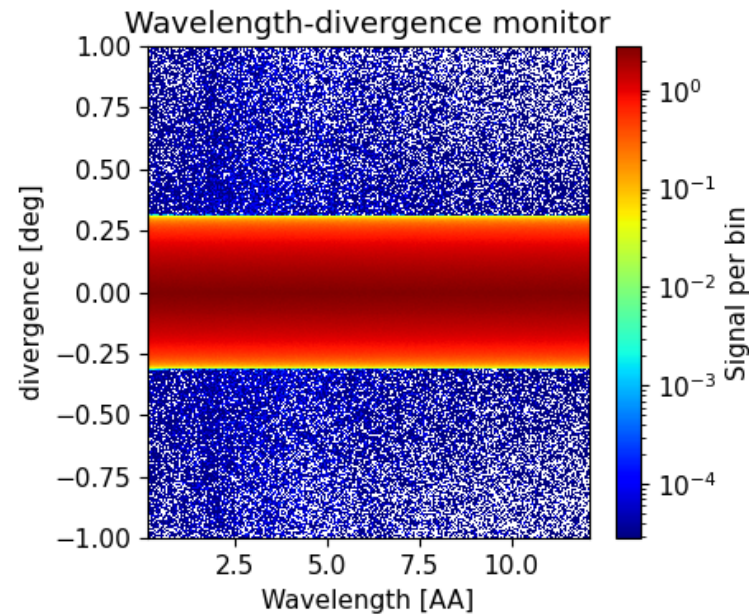
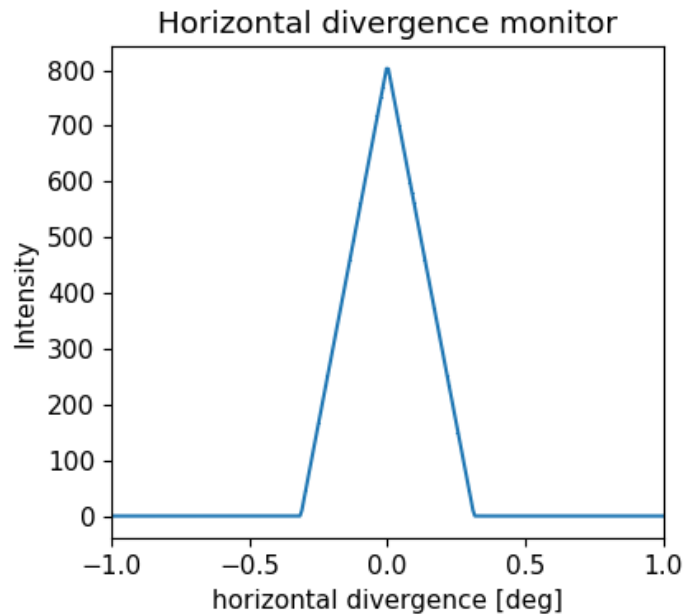
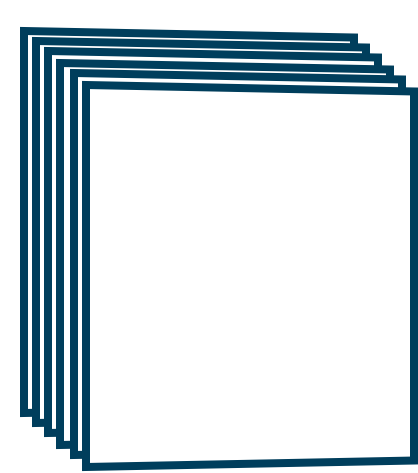
Union components

Simple example - colimator

- Lets make a colimator
- Absorbing blades limit divergence

NCrystal

McStas



Cd

Air

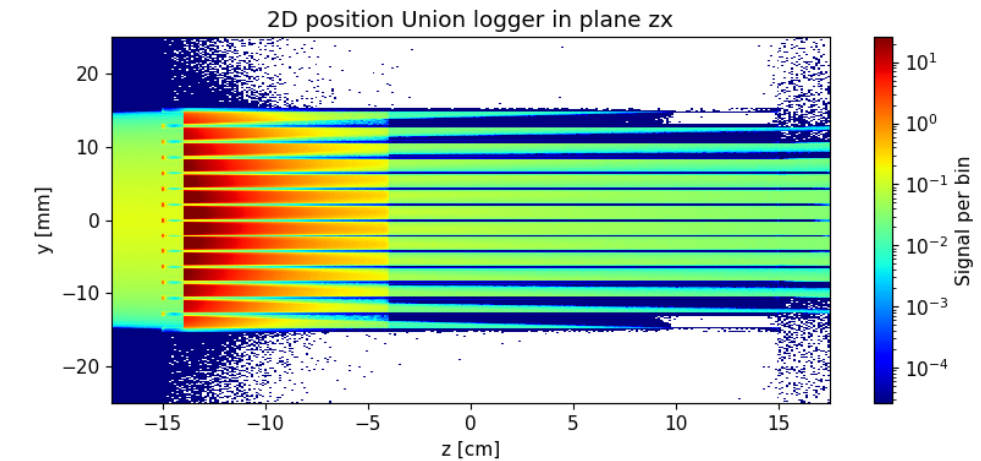
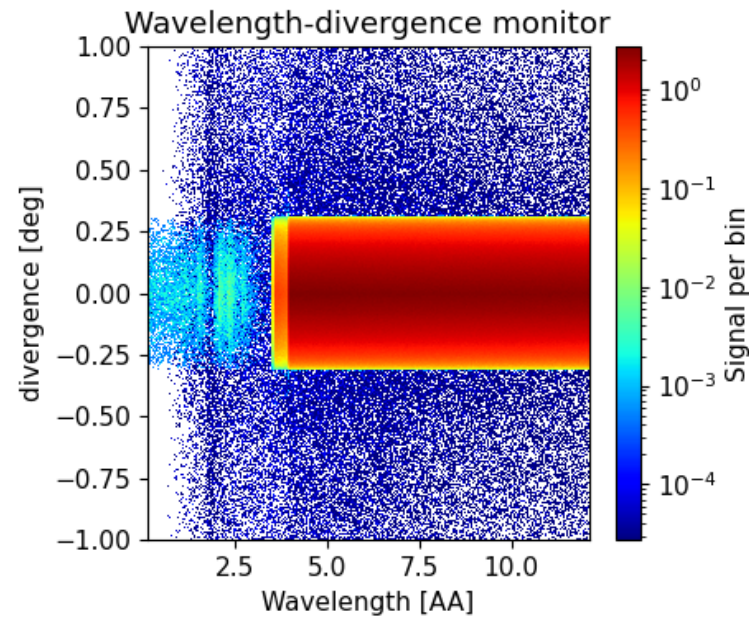
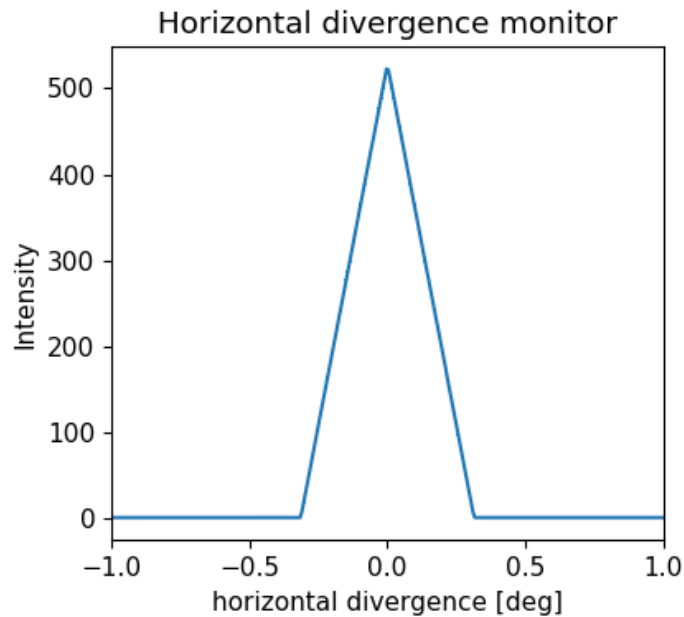
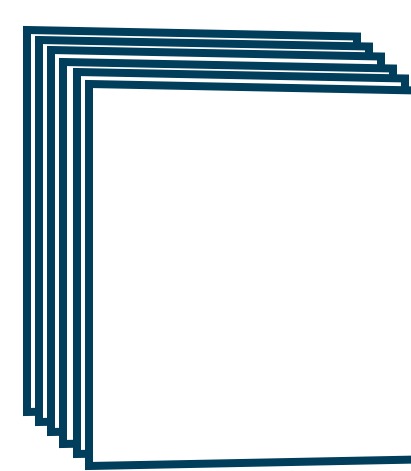
Union components

Simple example - filter

- Lets make a colimator
- Absorbing blades limit divergence
- Add a Be block, now its a filter

NCrystal

McStas



Cd

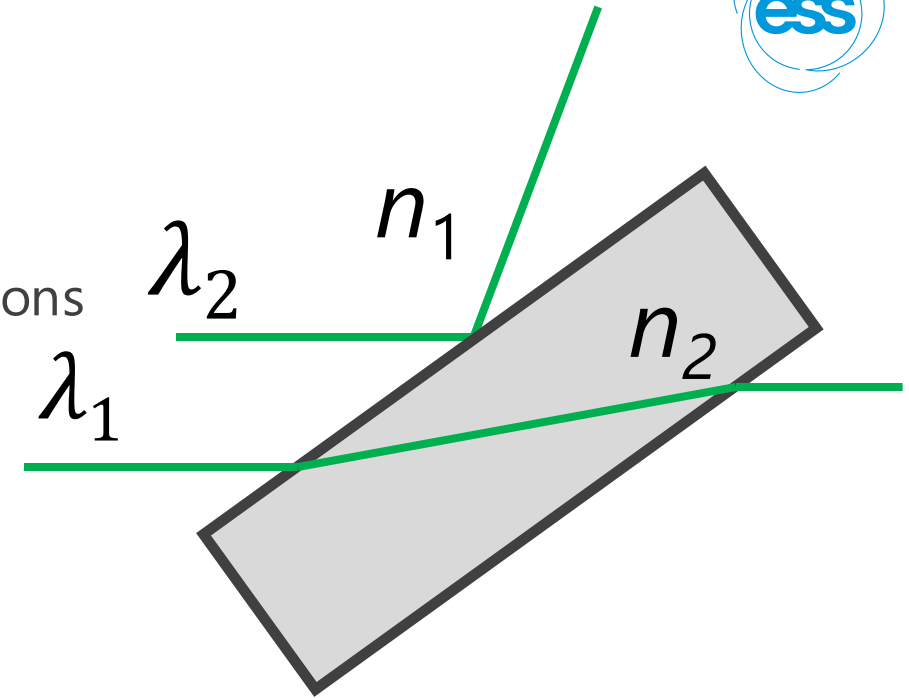
Air

Be

Interface physics update

Refraction and reflection

- Code from Refractor.comp by E. Farhi, B. Cubitt
- Scattering length density provided by material definitions
- Refraction require SLD for both materials
- Snells law with fall of
- Refraction / Reflection can be disabled in master with
 - enable_refraction = 0
 - enable_reflection = 0



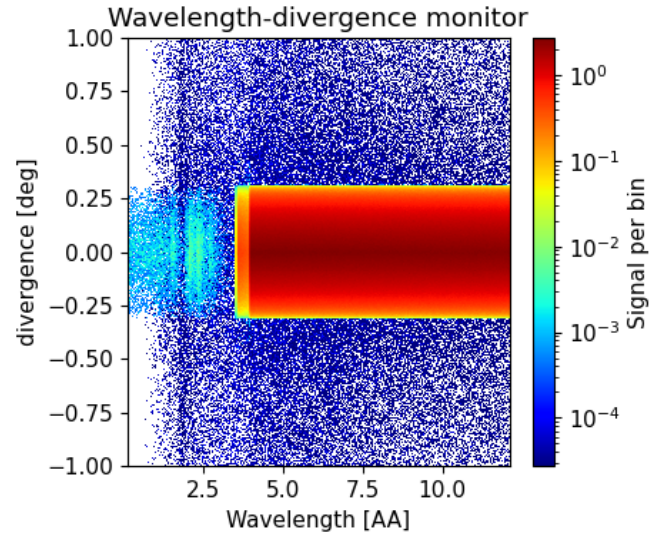
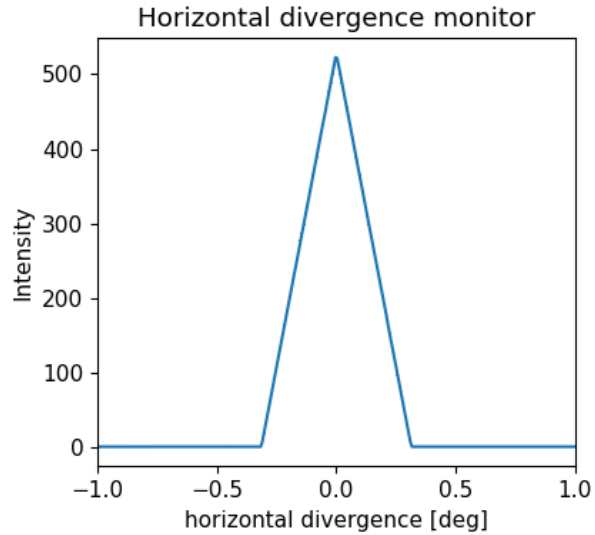
$$n = \sqrt{1.0 - (\lambda * \lambda * SLD / \pi)}$$

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$

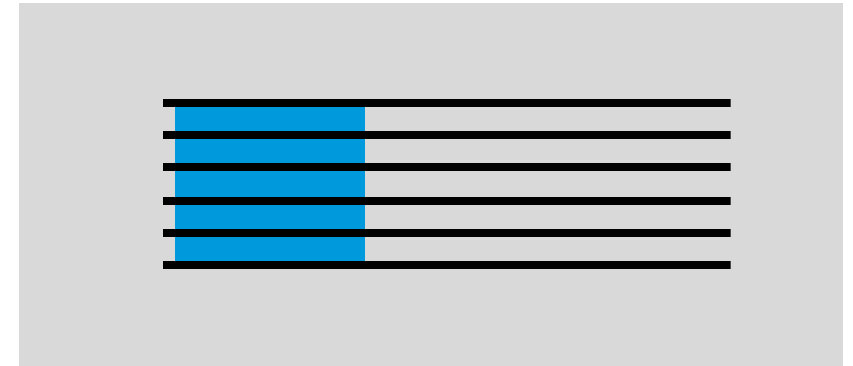
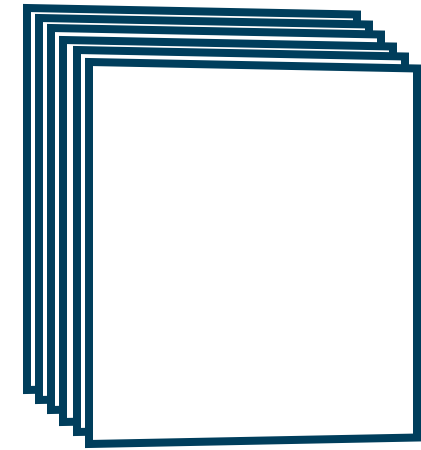
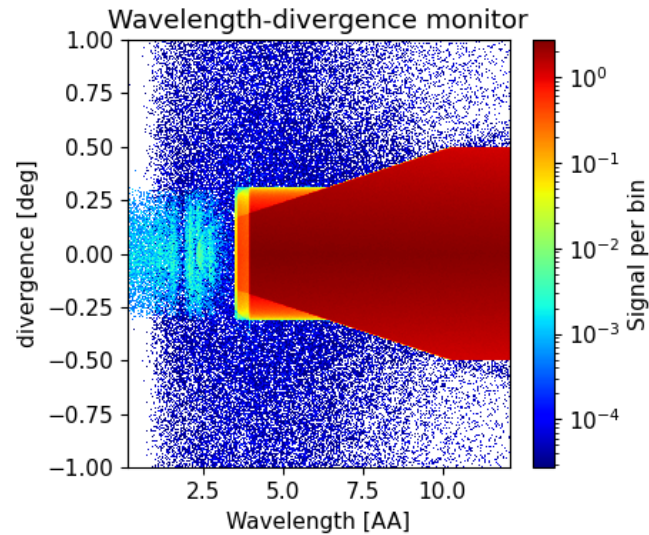
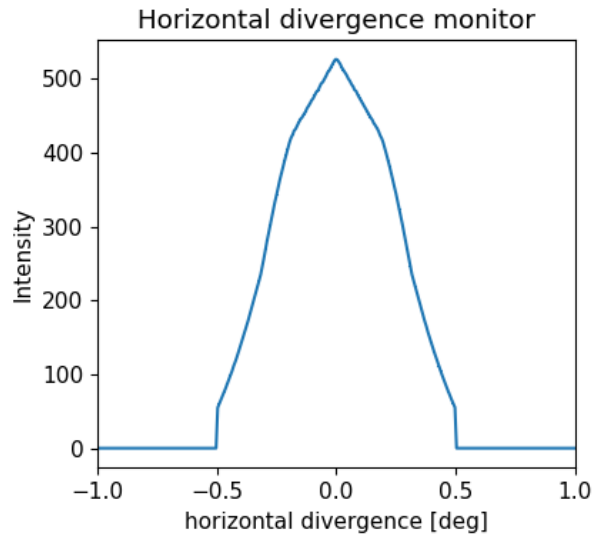
Interface physics update

Simple example - filter

No refraction



Refraction



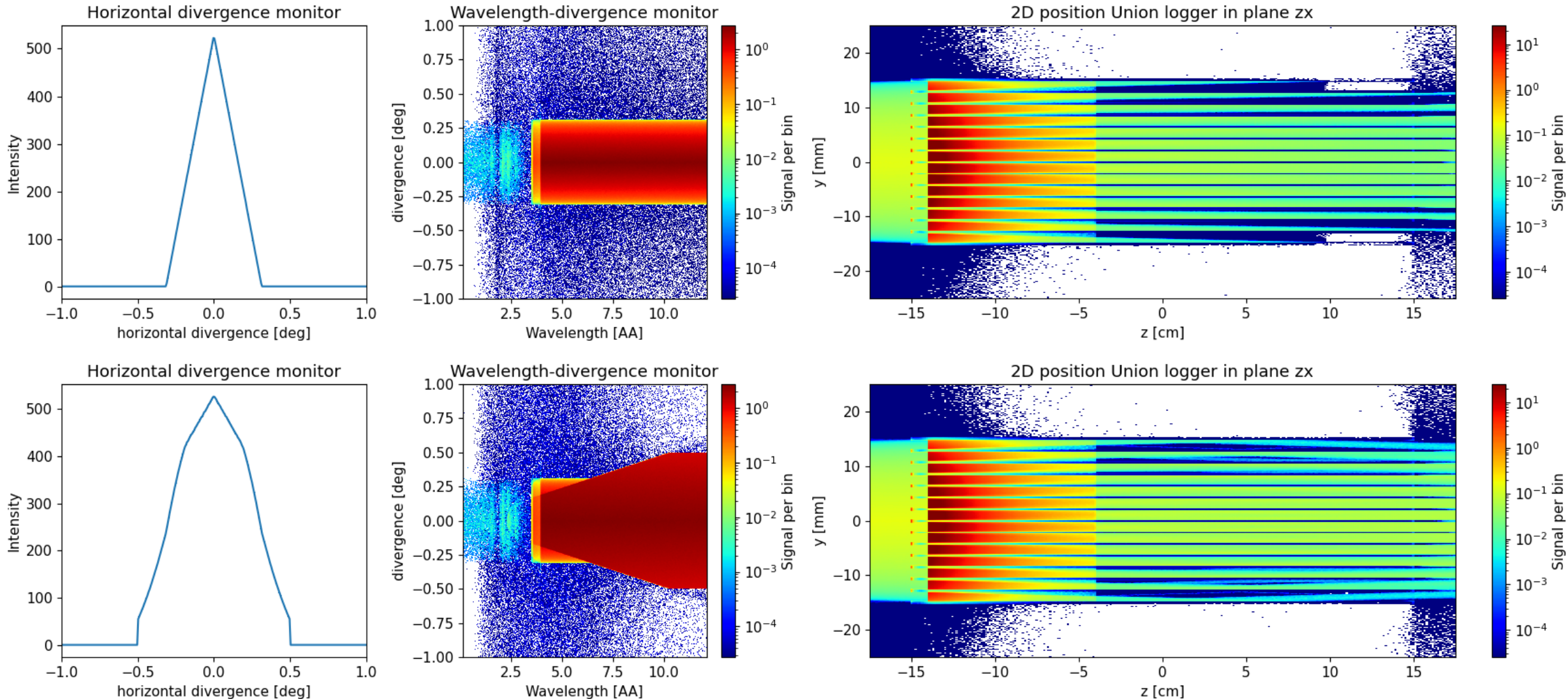
Cd

Air

Be

Interface physics update

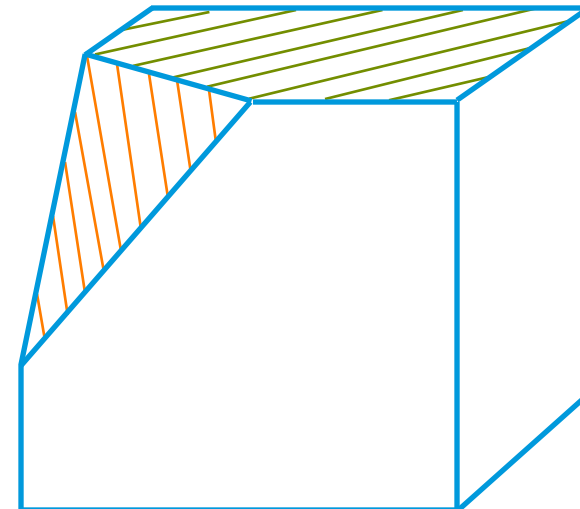
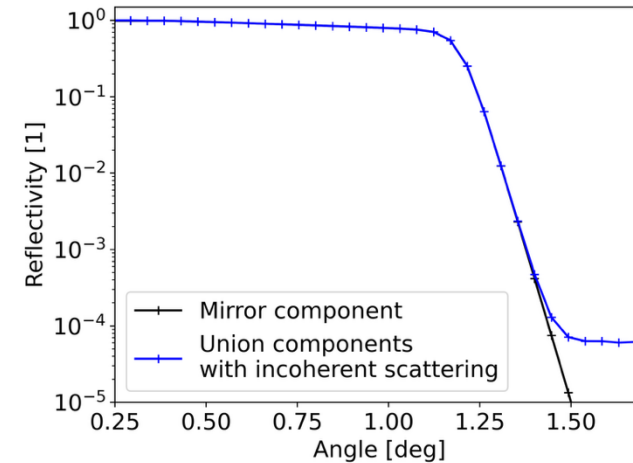
Simple example - filter



Interface physics update

Surface definitions

- Surface process components
 - Gets normal vector, initial wavevector and weight
 - Returns final wavevector and weight
- Allows
 - Reflectivity curve
 - Thin absorbing paints
 - ...
- Can be attached to geometry faces
 - Each outside face
 - ... and one for all cuts in inside of material



Interface physics update

Surface definitions

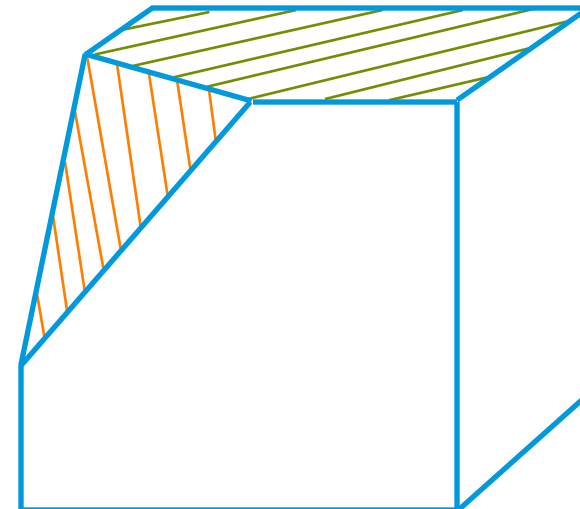
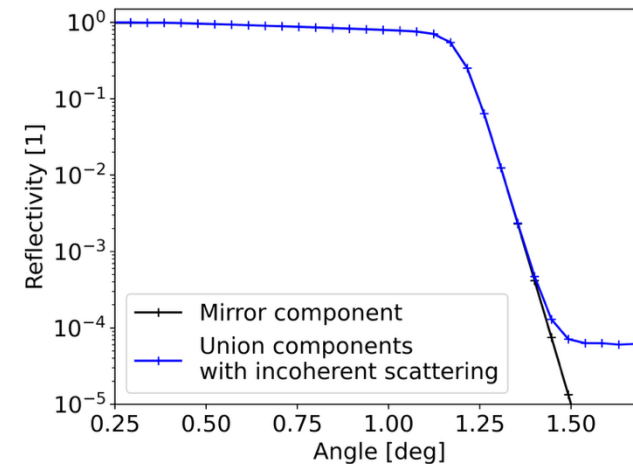
```
COMPONENT mirror_m4 = Mirror_process(m=4, alpha=3.8)
AT (0,0,0) ABSOLUTE
```

```
COMPONENT reflectometry_sample = Union_box(
  xwidth = 0.02, yheight = 0.01, zdepth = 0.08,
  priority = 10, material_string = "Al",
  plus_y_surface = "mirror_m4")
AT (0,0,0) RELATIVE sample_position
```

All geometries have the parameters:

- all_face_surface
- cut_surface

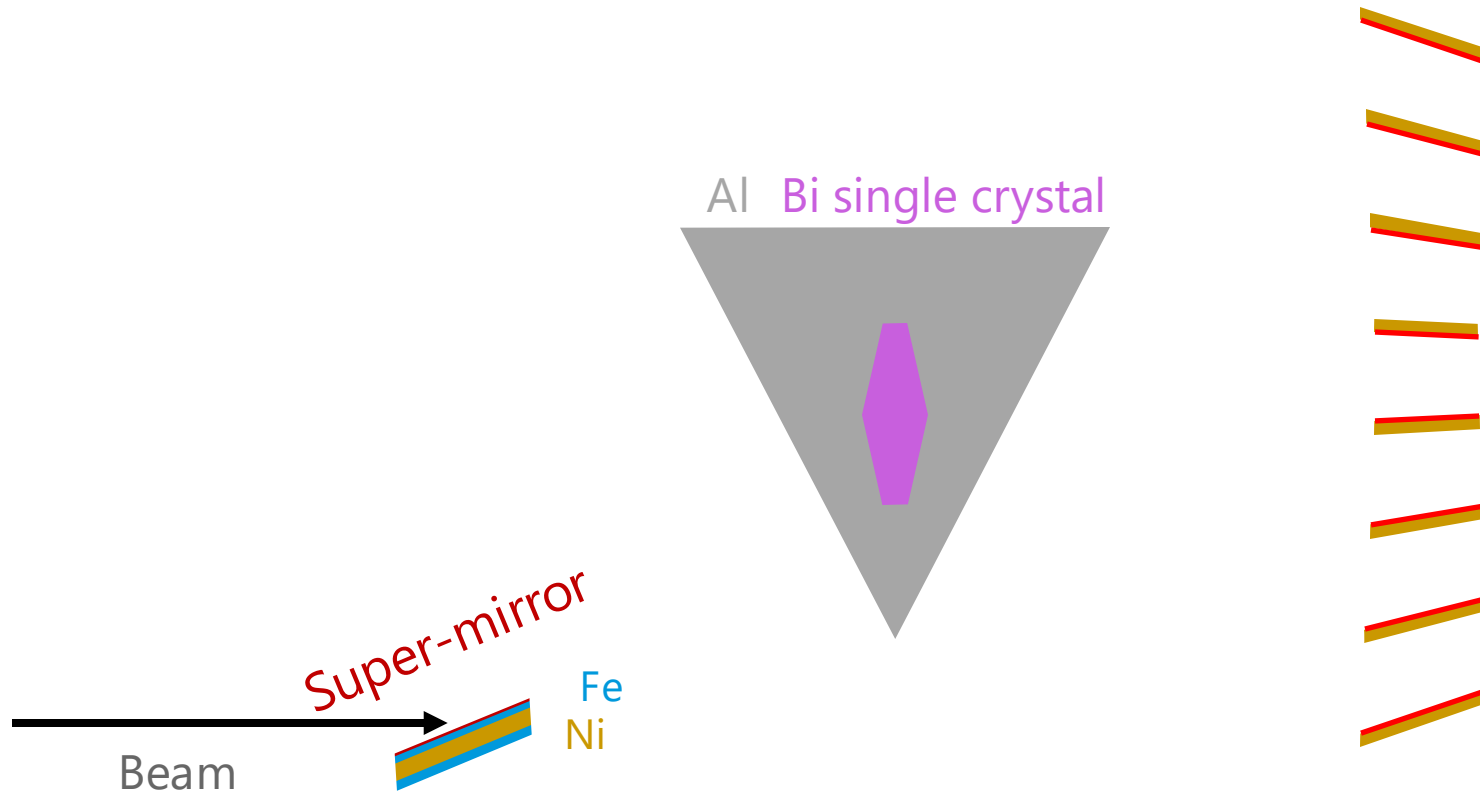
And then additional for the individual faces



Interface physics demo

NCrystal

McStas
 n



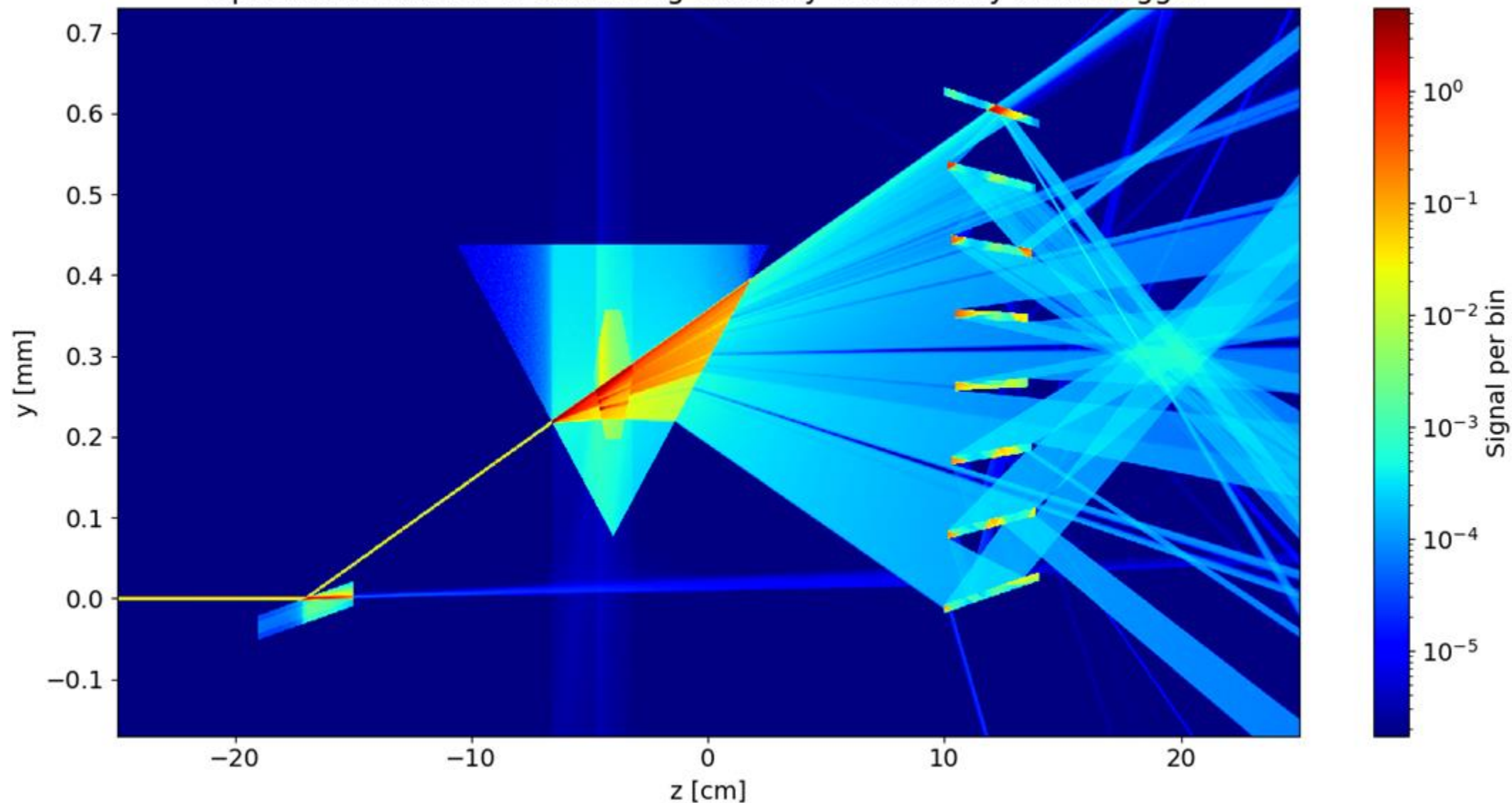
Interface physics demo

NCrystal

McStas
 n



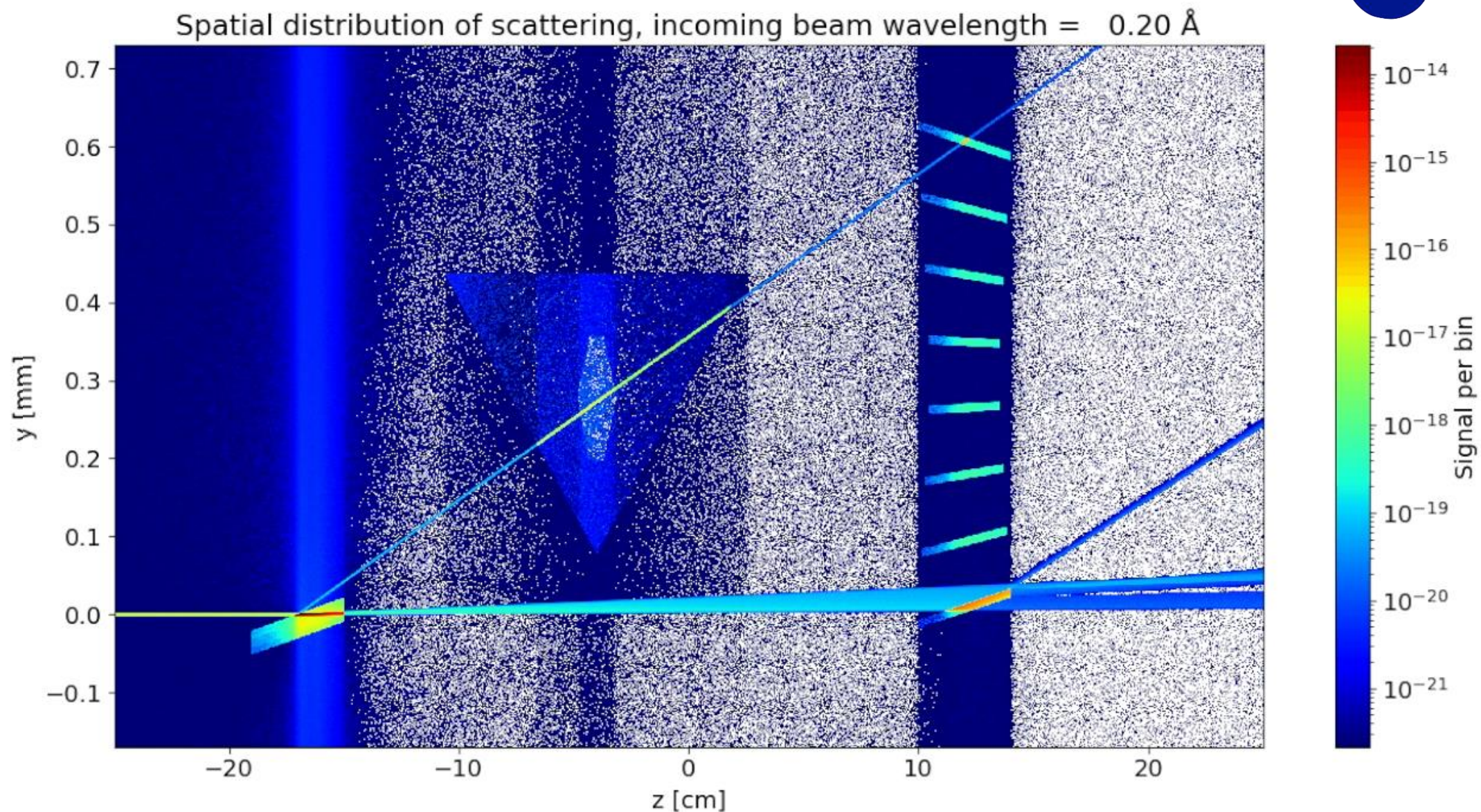
Spatial distribution of scattering intensity recorded by Union logger



Interface physics demo

NCrystal

McStas





Summary

- Union components allow simulation of complex physics and geometry
- Assumptions: No gravity, uniform materials, no polarization
- Full multiple scattering simulation provides surprising level of detail
- The modular nature makes it easier to expand
 - New geometry with a geometry component (Master project contributed cone and CAD)
 - New process component to add new physics
- Now supports refraction / reflection and surface physics such as mirrors
- Full tutorial available in McStasScript format!

Exercise

Build an interesting Union system

- An instrument file is provided with:
 - A number of Union processes and material definitions
 - A source, transmission detector and scattering detector
 - A logger that shows spatial distribution of scattering seen from above (zx plane)
 - An absorption logger that shows spatial distribution of absorption seen from above

- Your task is to use geometry components to build an interesting sample
 - Remember to give all geometries a priority, higher priority wins in overlaps
 - Perhaps start adding a sample environment
 - Send me pictures! mads.bertelsen@ess.eu



Thanks for your attention